

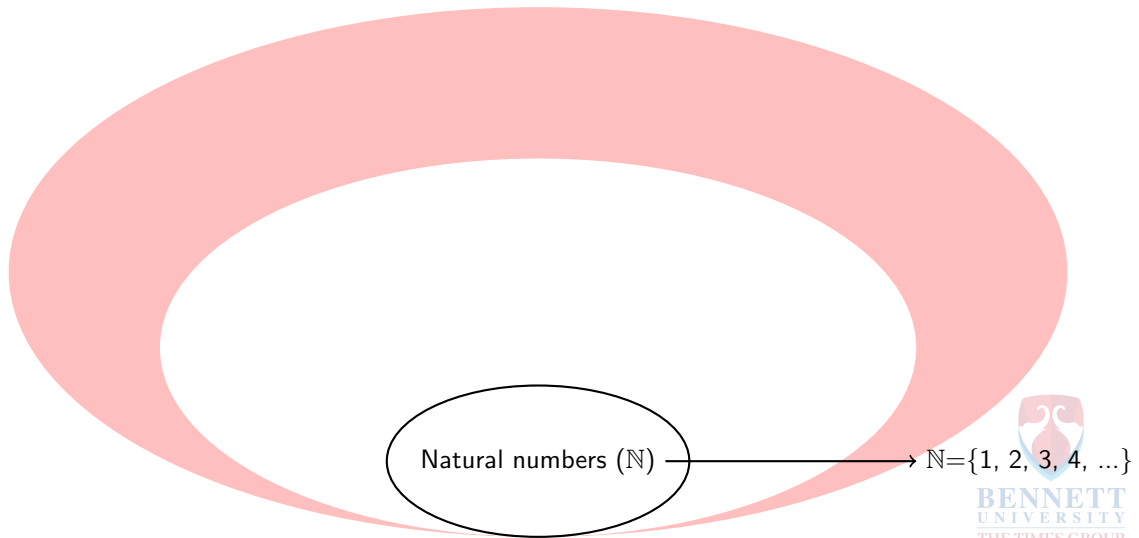
Engineering Calculus EMAT101L

Lecture 2

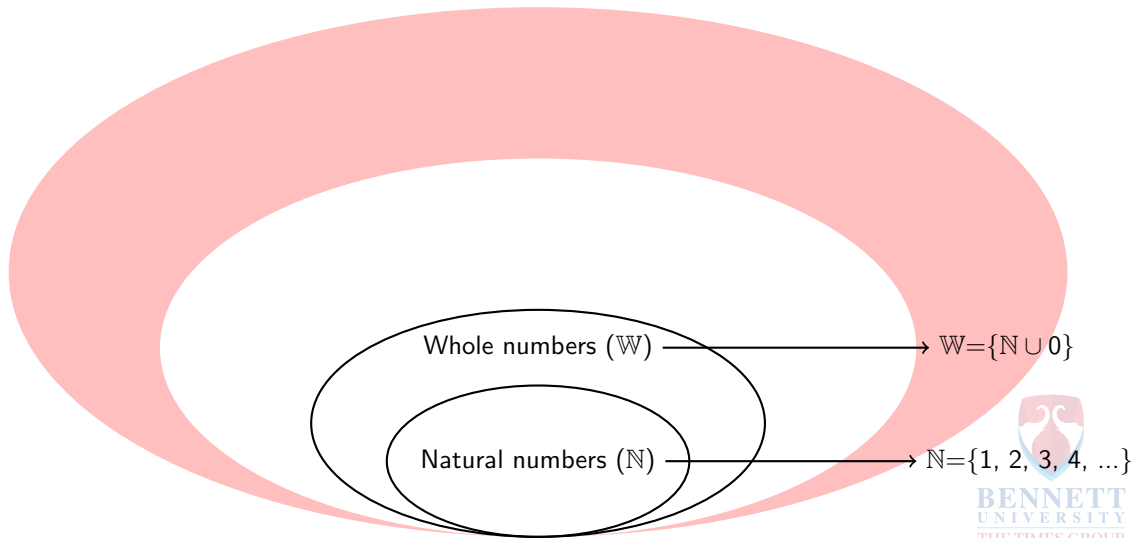
Bennett University

- The Real number system,
- Archimedean property,
- Sequence and its convergence and divergence,
- Monotone sequences,
- Cauchy Criterion,
- Bolzano-Weierstrass Theorem,
- Limit inferior and limit superior.
- Infinite Series and its convergence,
- Alternating series,
- Limit and continuity,
- Existence of maxima & minima,
- Intermediate Value Property.

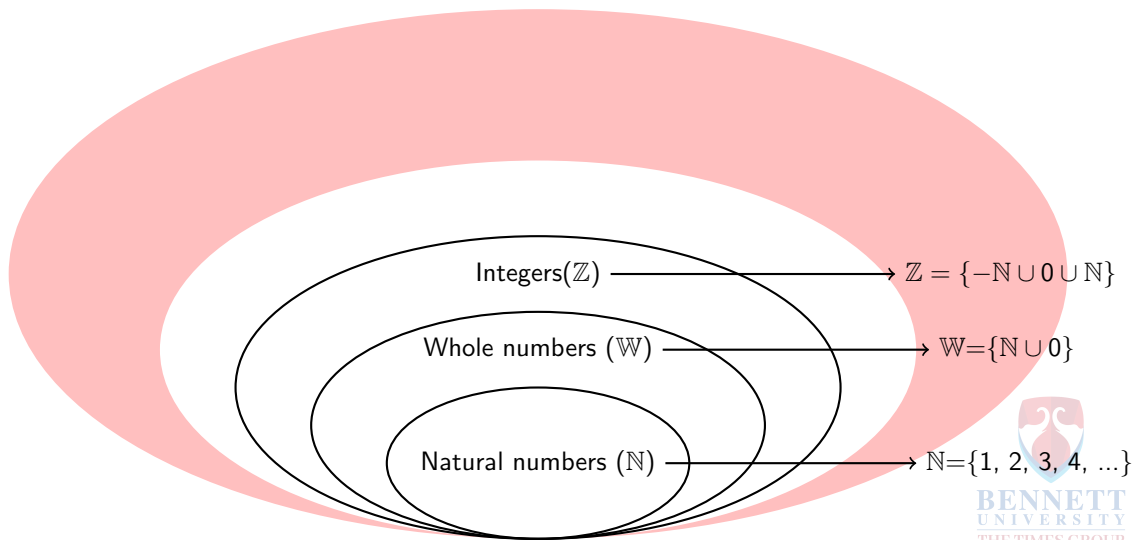
The real number system



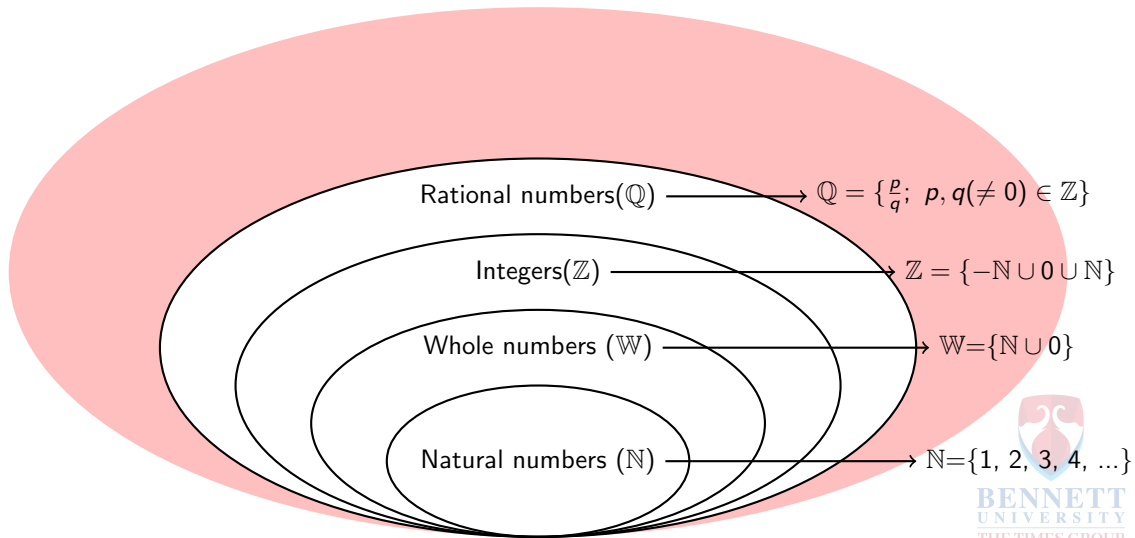
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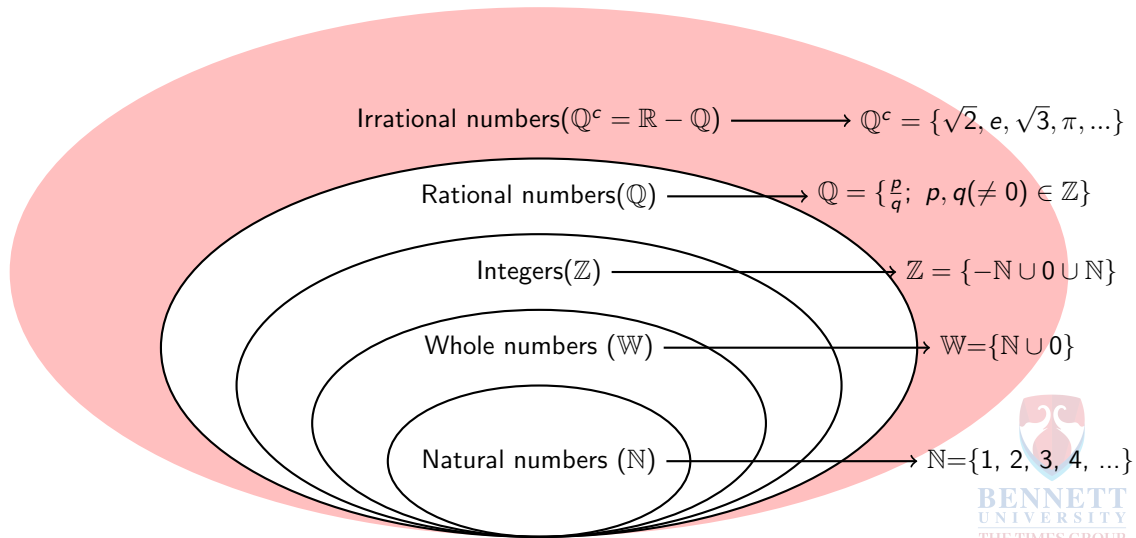
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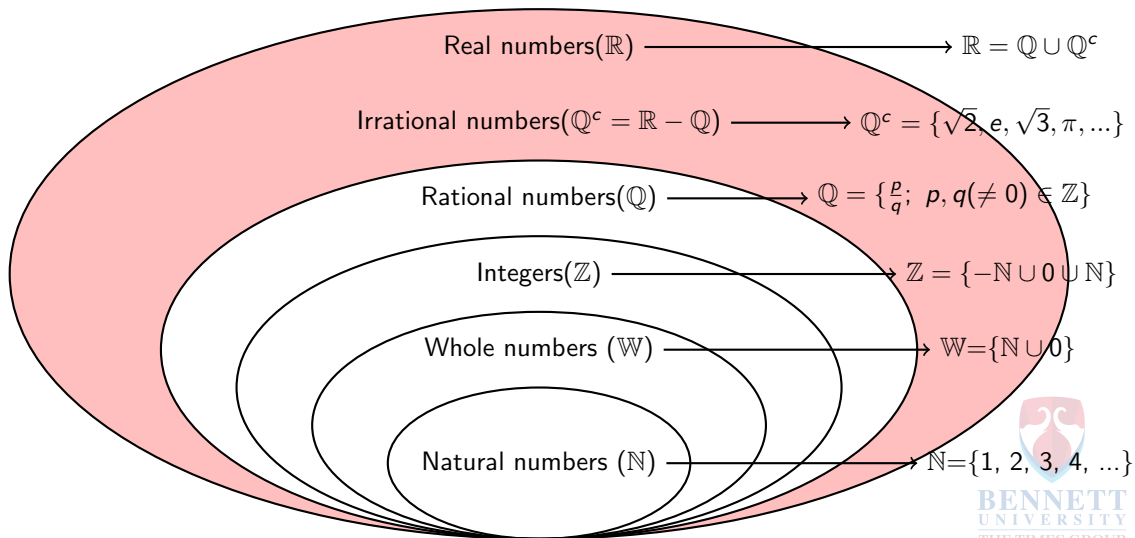
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The real number system



The real number system



1 2

Examples

- 1 2
- 2 2.5

Examples

- ① 2
- ② 2.5
- ③ π

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③ $\pi = 3.141592653589793238462643383279502884197...$

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Archimedean Property

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Archimedean Property

If x and y are any two positive real numbers with $y < x$, then there exists a positive integer n such that $ny > x$.

The most convenient and familiar way to represent real number system is the real line $\mathbb{R}$

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Real line



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Problem 1 What is the distance between points 0.2 and 0.8.

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Each number 'a' on the line represents the distance from the origin, i.e., 0.

Problem 1 What is the distance between points 0.2 and 0.8.

Problem 2 : What is the distance between points 0.2 and -0.8 .

A precise formal way of saying a real number 'x' is "close to" another point 'a' is to say that the distance $|x - a|$ is sufficiently small, i.e.,

$$|x - a| < \epsilon \quad \text{for} \quad \epsilon > 0,$$

here ϵ is "very small" quantity near to zero.

The term "very small" is relative and it depends on what scaling we are using:

e.g., (i) distance between two colliding stars.

(ii) thickness of a paper.

(iii) distance between two blood cells.

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Definition

Let $a \in \mathbb{R}$ and $\epsilon > 0$. The ϵ -**neighbourhood** of a is defined as:

$$N_{\epsilon}(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\} = \{x \in \mathbb{R} : x \in (a - \epsilon, a + \epsilon)\}$$

Open interval

$$(a, b) = \{x \in \mathbb{R} : a < x < b, a, b \in \mathbb{R}\}$$

Intervals

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$$[a, b) = \{x \in \mathbb{R} : a \leq x < b, a, b \in \mathbb{R}\} \text{ and } (a, b] = \{x \in \mathbb{R} : a < x \leq b, a, b \in \mathbb{R}\}$$

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Length of intervals in all above cases is $b - a$. If $a = b$ then

Q1: $[a, a] =$

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Bounds

Let S be a non-empty set.

Maximum of a set

The largest element, M , of the set S (if exists) is called the maximum of the S .

Minimum of a set

The smallest element, m , of the set S (if exists) is called the minimum of the S .

Problems:

Q1: If $S = [1, 2]$, what is $\max(S)$ and $\min(S)$?

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Q4: If $S = \{1, 1/2, 1/3, 1/4, \dots\}$, what is $\max(S)$ and $\min(S)$?

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Ans: 1, There is no minimum as for all $1/n \in S$ the number $1/(n+1)$ is in the set and smaller..

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A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded above** if there is an element $K \in \mathbb{R}$ such that $x \leq K$ for all $x \in S$. The number K is said to be **upper bound** of S .

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Bounded set

A non-empty set $S \subseteq \mathbb{R}$ is said to be **bounded** if it is bounded above as well as bounded below.

Ex: $S = \{x \in \mathbb{R} : x < 2\}$. Then S is

(a) Bounded, (b) Bounded above, (c) Bounded below, (d) Unbounded.?

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Answer: (b) and (d).

In fact, every number greater than 2 is also an upper bound of S .

Ex: If $S = (1, 2)$, then

- Upper bounds of $S = [2, \infty)$ and
- Lower bounds of $S = (-\infty, 1]$.
- S is bounded.
- All results hold same for $S = [1, 2]$.

Ex: Find the bounds of $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$.

Supremum & Infimum

Let $S(\neq \emptyset) \subseteq \mathbb{R}$.

Supremum (or least upper bound)

If S is bounded above, then the **smallest number** K among all the upper bounds of S is said to be the supremum of S . Denoted by $\sup(S)$.

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Example: $S = [1, 3, 5, 7, 9]$.
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Q1: $S = \{x : x > 0, x \in \mathbb{R}\}$. Find $\inf(S)$, and $\sup(S)$.

Examples

Q1: $S = \{x : x > 0, x \in \mathbb{R}\}$. Find $\inf(S)$, and $\sup(S)$.

Ans: $\inf(S) = 0$ and $\sup(s)$ do not exist. Also, infimum do not belong to the set.

Q2: The infinite set $S = \{x : 0 \leq x \leq 1, x \in \mathbb{Q}\}$. Is this set bounded?

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Ans: The set S is bounded.

$\sup(S) = 1$ and $\inf(S) = 0$, both belongs to the set.

Q3: Consider $S = \{\frac{1}{n} : n \in \mathbb{N}\}$. Is this bounded?

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Ans: The set S is bounded. $\sup(S) = 1$, $\sup(S) \in S$.

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Ans: The set S is bounded.

$\sup(S) = 1$ and $\inf(S) = 0$, both belongs to the set.

Q3: Consider $S = \{\frac{1}{n} : n \in \mathbb{N}\}$. Is this bounded?

Ans: The set S is bounded. $\sup(S) = 1$, $\sup(S) \in S$.

$\inf(S) = 0$,

Examples

Q1: $S = \{x : x > 0, x \in \mathbb{R}\}$. Find $\inf(S)$, and $\sup(S)$.

Ans: $\inf(S) = 0$ and $\sup(s)$ do not exist. Also, infimum do not belong to the set.

Q2: The infinite set $S = \{x : 0 \leq x \leq 1, x \in \mathbb{Q}\}$. Is this set bounded?

Ans: The set S is bounded.

$\sup(S) = 1$ and $\inf(S) = 0$, both belongs to the set.

Q3: Consider $S = \{\frac{1}{n} : n \in \mathbb{N}\}$. Is this bounded?

Ans: The set S is bounded. $\sup(S) = 1$, $\sup(S) \in S$.

$\inf(S) = 0$, $\inf(S) \notin S$.

Q4: Discuss the bounds of: (i) $[a,b]$, (ii) $(a,b]$, (iii) $[a,b)$, (iv) (a,b) .

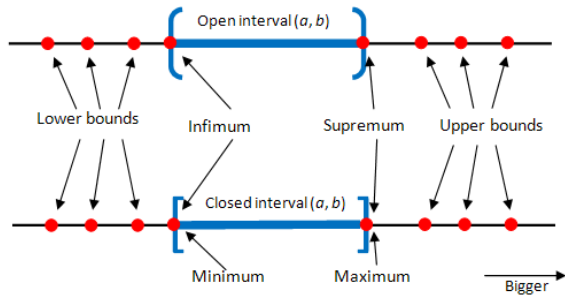


Maxima/Minima, Lower/Upper bounds, Supremum/Infimum

Note 1: Upper bound and Lower bound **may not belong** to the set.

Note 2: Maximum and minimum **always belong** to the set.

Note 3: Supremum and Infimum **may not belong** to the set.



Archimedian property

Let x be any real number. Then there exists a natural number n such that $n > x$.